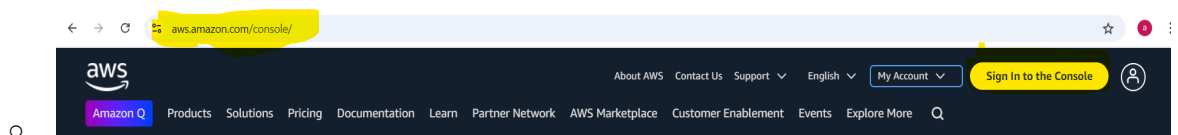


Sure! Here's a detailed explanation of how to perform the steps you've described using **Elastic Block Store (EBS)** on AWS. This involves creating an EC2 instance, creating and attaching an EBS volume, taking a snapshot, and then creating a new volume from that snapshot to attach to another instance.

Step 1: Create an EC2 Instance

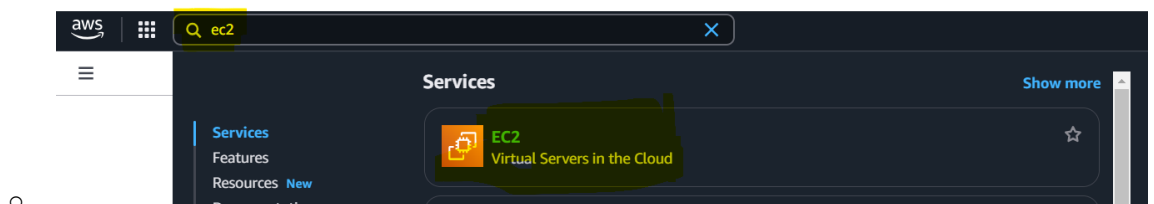
1. Sign in to AWS Console:

- Go to the [AWS Management Console](https://aws.amazon.com/console/) and sign in.

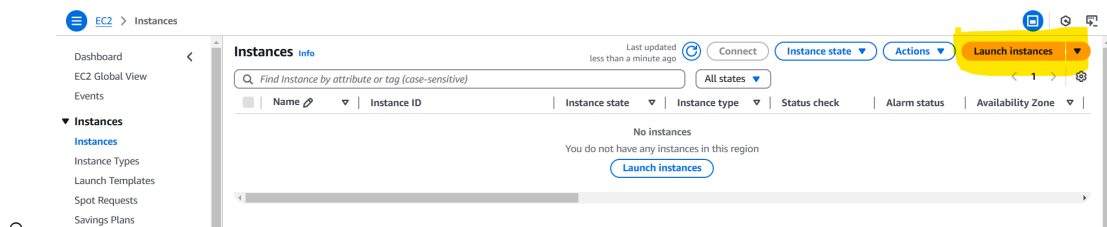


2. Launch EC2 Instance:

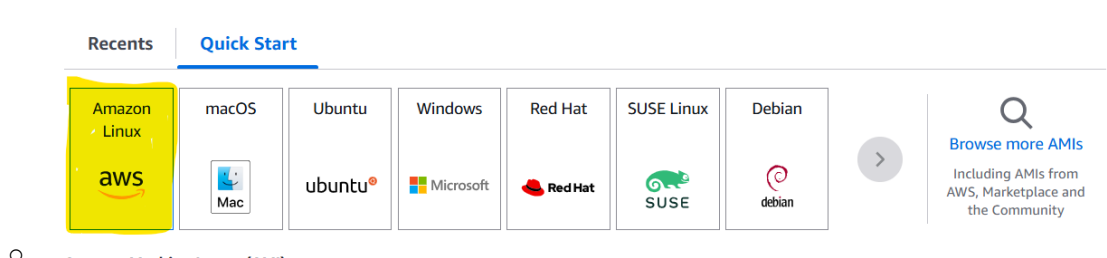
- Navigate to the **EC2 Dashboard**.



- Click **Launch Instance** to create a new instance.



- Choose an Amazon Machine Image (AMI) (e.g., Amazon Linux, Ubuntu).



- Select the **Instance Type** (e.g., t2.micro for testing).



- Configure **Instance Details**, such as network and IAM roles (you can leave the defaults if you're just testing).
- **Add Storage**: Ensure there's an EBS volume attached (this is the root volume by default).

▼ **Configure storage** [Info](#) Advanced

1x 8 GiB gp3 — Root volume 3000 IOPS (Not encrypted)

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[Add new volume](#)

-
- **Configure Security Group**: Set appropriate rules (e.g., allow SSH access on port 22).

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-3' with the following rules:

☒ Allow SSH traffic from
 Helps you connect to your instance
 Anywhere
 0.0.0.0/0

☐ Allow HTTPS traffic from the internet
 To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet
 To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

-
- Review and **Launch** the instance. Choose or create a key pair to access the instance via SSH.

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

virginia

[Create new key pair](#)

3. Connect to the Instance:

- After the instance is running, you can connect to it using an SSH client or AWS EC2 Instance Connect (for browser-based SSH).

○

○

Step 2: Create an EBS Volume

1. Go to the EC2 Dashboard:

- In the **Elastic Block Store** section, click **Volumes**.

○

○

○

2. Configure the Volume:

- **Volume Type:** Select the volume type (e.g., **General Purpose SSD (gp3)**).

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp3)

○

- **Size:** Specify the size (e.g., 10 GiB).

EC2 > Volumes > Create volume

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp3)

Size (GiB) [Info](#)

100

Min: 1 GiB, Max: 16384 GiB.

○

- **Availability Zone:** Choose the same availability zone as your EC2 instance to ensure the volume can be attached to it.

Throughput (MiB/s) [Info](#)

125

Min: 125 MiB, Max: 1000 MiB. Baseline: 125 MiB/s.

Availability Zone [Info](#)

us-east-1c

○

- Leave the other options as default unless you need encryption or specific settings.

3. Create Volume:

- After configuring, click **Create Volume**.

EC2 > Volumes > Create volume

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add tag](#)

You can add 50 more tags.

Snapshot summary [Info](#)

Click refresh to view backup information

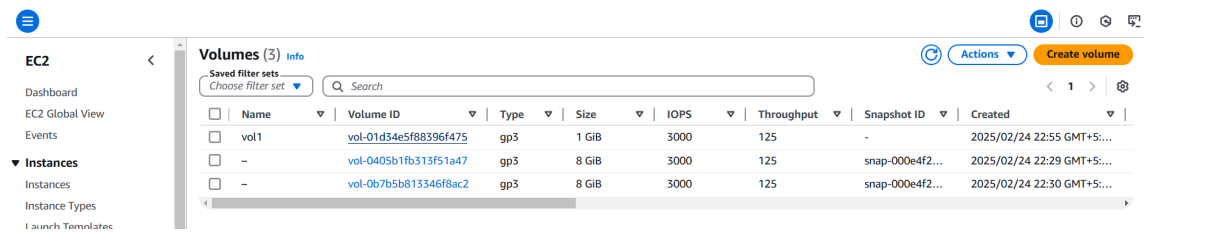
The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

[Cancel](#)

[Create volume](#)

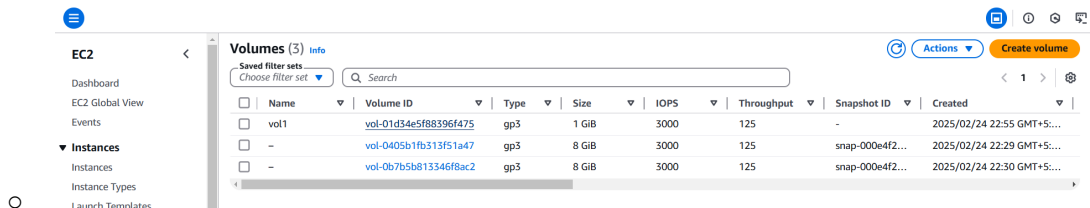
○

Step 3: Attach the EBS Volume to the EC2 Instance

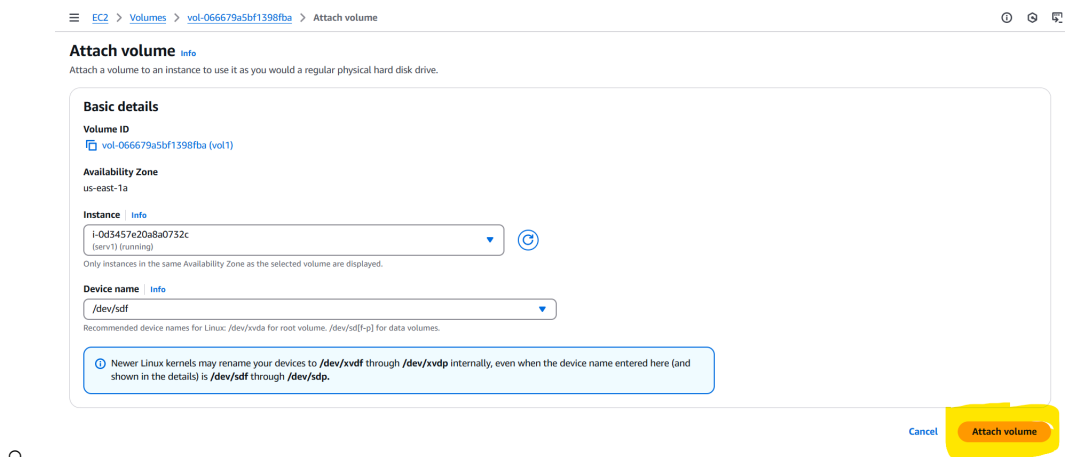


1. Attach Volume:

- Once the volume is created, go to the **Volumes** page.



- Select the newly created volume and click **Actions > Attach Volume**.



2. Select EC2 Instance:

- Choose the EC2 instance you want to attach it to (the instance you created in Step 1).
- Choose a device name (e.g., `/dev/xvdf` or `/dev/sdf`).

3. Attach Volume:

- Click **Attach** to attach the volume to the instance.

Step 4: Format and Mount the EBS Volume on the EC2 Instance

1. Connect to EC2:

- Use SSH to log into the instance.

EC2 > Instances > i-Od3457e20a8a0732c > Connect to instance

Connect to instance [info](#)

Connect to your instance i-Od3457e20a8a0732c (serv1) using any of these options

EC2 Instance Connect Session Manager **SSH client** EC2 serial console

Instance ID
i-Od3457e20a8a0732c (serv1)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is `virginia.pem`.
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "virginia.pem"`
4. Connect to your instance using its Public DNS:
`ec2-52-23-250-150.compute-1.amazonaws.com`

Command copied

```
ssh -i "virginia.pem" ec2-user@ec2-52-23-250-150.compute-1.amazonaws.com
```

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

2. Format the Volume:

- Run the following command to list available disks:
- `lsblk`

```
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0   8G  0 disk
├─xvda1     202:1    0   8G  0 part /
├─xvda127   259:0    0    1M  0 part
└─xvda128   259:1    0   10M  0 part /boot/efi
```

- Identify the new volume (e.g., `/dev/xvdf`), then format it using:
- `sudo mkfs -t ext4 /dev/xvdf`

```
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0   8G  0 disk
├─xvda1     202:1    0   8G  0 part /
├─xvda127   259:0    0    1M  0 part
└─xvda128   259:1    0   10M  0 part /boot/efi
xvdf        202:80    0    1G  0 disk
[root@serv1 ~]#
[root@serv1 ~]# df -Th
Filesystem      Type      Size  Used Avail Use% Mounted on
devtmpfs        devtmpfs  4.0M   0    4.0M   0% /dev
tmpfs           tmpfs     475M   0    475M   0% /dev/shm
tmpfs           tmpfs     190M  452K   190M   1% /run
/dev/xvda1      xfs       8.0G  1.6G   6.4G  20% /
tmpfs           tmpfs     475M   0    475M   0% /tmp
/dev/xvda128    vfat      10M   1.3M   8.7M  13% /boot/efi
tmpfs           tmpfs     95M   0     95M   0% /run/user/1000
```

3. Create a Mount Point:

- Create a directory where the volume will be mounted:
- `sudo mkdir /data`

4. Mount the Volume:

- Mount the volume to the new directory:
- `sudo mount /dev/xvdf /data`

```

root@serv1 ~]# mount -t ext4 /dev/xvdf /asha
root@serv1 ~]#
root@serv1 ~]# cd /asha
root@serv1 asha]#
root@serv1 asha]# touch file1 file2 file3
root@serv1 asha]# mkdir dir1 dir2
root@serv1 asha]#
root@serv1 asha]# ll
total 24
drwxr-xr-x. 2 root root 4096 Feb 24 18:13 dir1
drwxr-xr-x. 2 root root 4096 Feb 24 18:13 dir2
-rw-r--r--. 1 root root    0 Feb 24 18:12 file1
-rw-r--r--. 1 root root    0 Feb 24 18:12 file2
-rw-r--r--. 1 root root    0 Feb 24 18:12 file3
drwx-----. 2 root root 16384 Feb 24 18:09 lost+found

```

Step 5: Create a Snapshot of the Volume

1. Take Snapshot:

- In the **Volumes** section of the EC2 dashboard, select the volume.

The screenshot shows the AWS Management Console's EC2 dashboard. On the left sidebar, under the 'Elastic Block Store' section, the 'Volumes' link is highlighted in yellow. The main content area displays a summary of EC2 resources in the 'United States (N. Virginia)' region, including instances, auto scaling groups, capacity reservations, dedicated hosts, elastic IPs, key pairs, load balancers, placement groups, security groups, snapshots, and volumes. A 'Launch instance' button is visible. On the right, there's a section for 'EC2 Free Tier' offers and 'Offer usage (monthly)' for Linux EC2 instances and storage space on EBS.

- Click **Actions > Create Snapshot**.

The screenshot shows the 'Volumes' page in the AWS Management Console. A green banner at the top indicates 'Successfully deleted volume vol-01d34e5f88396f475'. Below this, a table lists three volumes. The volume 'vol1' (ID: vol-066679a5bf1398fba) is selected, and a context menu is open over it. The 'Create snapshot' option in the menu is highlighted in yellow. The details section for the selected volume shows its ID, name, type (gp3), size (1 GiB), IOPS (3000), and throughput (125). The 'Actions' button in the top right corner is also visible.

- Add a **Name** and **Description** for the snapshot and click **Create Snapshot**.

Snapshot details

Description

Add a description for your snapshot

snapshot-1

255 characters maximum.

Encryption

info

Not encrypted

Tags

info

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add tag

You can add 50 more tags.

Cancel

Create snapshot

2. Wait for Snapshot to Complete:

- The snapshot will be available in the **Snapshots** section once it's completed. This snapshot is essentially a backup of your volume.

Step 6: Create a New EBS Volume from the Snapshot

1. Go to Snapshots:

- In the EC2 dashboard, click on **Snapshots** under **Elastic Block Store**.

EC2

Dashboard

EC2 Global View

Events

Instances

Images

Elastic Block Store

Snapshots

Resources

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)	2	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	4
Key pairs	6	Load balancers	0	Placement groups	0
Security groups	8	Snapshots	1	Volumes	3

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

Migrate a server

Note: Your instances will launch in the United States (N. Virginia) Region

Service health

AWS Health Dashboard

Region

United States (N. Virginia)

Status

This service is operating normally.

Zones

Zone name	Zone ID
-----------	---------

EC2 Free Tier

Offers for all AWS Regions.

2 EC2 free tier offers in use

End of month forecast

0 offers forecasted to exceed free tier limit.

Exceeds free tier

0 offers exceeded and is now pay-as-you-go pricing.

View Global EC2 resources

Offer usage (monthly)

Linux EC2 Instances

601.500835 hours remaining

20%

Storage space on EBS

26.55 GB remaining

12%

View all AWS Free Tier offers

Account attributes

2. Select Snapshot:

- Select the snapshot you created in Step 5.

EC2

Dashboard

EC2 Global View

Events

Instances

Snapshots (1/1)

info

Owned by me

Search

Recycle Bin

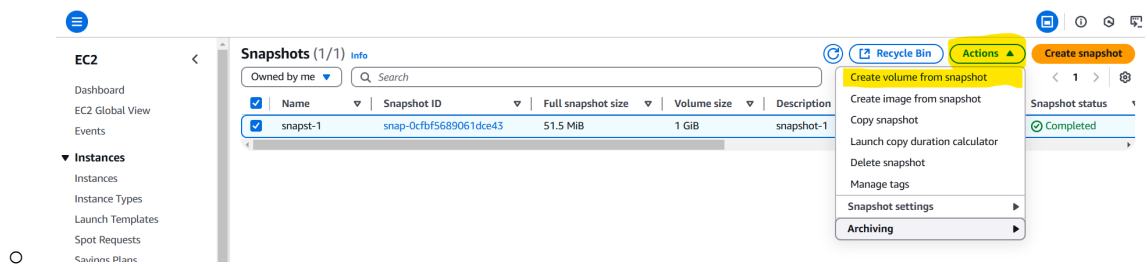
Actions

Create snapshot

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status
snaptst-1	snap-0c7bf5689061dce43	51.5 MiB	1 GiB	snapshot-1	Standard	Completed

3. Create Volume from Snapshot:

- Click **Actions > Create Volume**.



- Choose the **Volume Type**, **Size**, and **Availability Zone** for the new volume (make sure it's in the same zone as the EC2 instance where you plan to attach it).

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Snapshot ID
[snap-0cfbf5689061dce43 \(snapst-1\)](#)

Volume type [Info](#)
 General Purpose SSD (gp3)

Size (GiB) [Info](#)
 1
 Min: 1 GiB, Max: 16384 GiB.

IOPS [Info](#)
 3000
 Min: 3000 IOPS, Max: 16000 IOPS.

Throughput (MiB/s) [Info](#)
 125
 Min: 125 MiB, Max: 1000 MiB, Baseline: 125 MiB/s.

Availability Zone [Info](#)

Tags - optional [Info](#)
 A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
 No tags associated with the resource.
[Add tag](#)
 You can add 50 more tags.

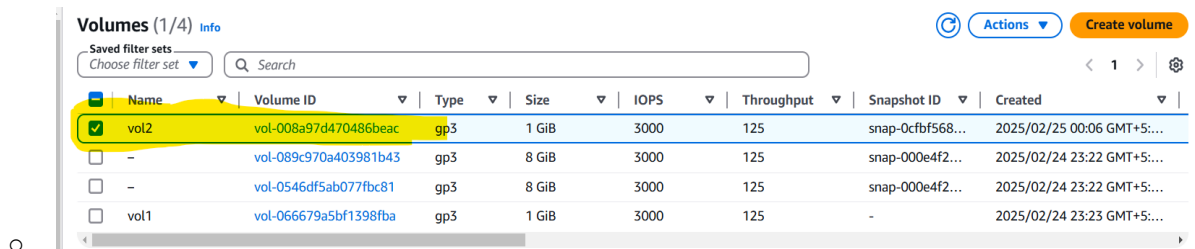
Snapshot summary [Info](#)
 Click refresh to view backup information
 The volume type that you select and the tags that you assign determine whether the volume will be backed up by any Data Lifecycle Manager policies.

[Cancel](#) [Create volume](#)

Step 7: Attach the New Volume to Another EC2 Instance

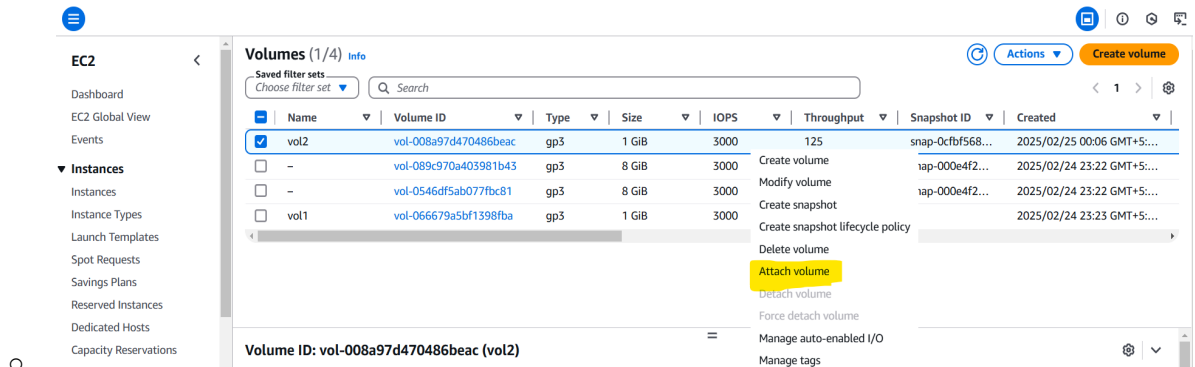
1. Select New Volume:

- After creating the new volume, go to the **Volumes** section.
- Select the volume created from the snapshot.

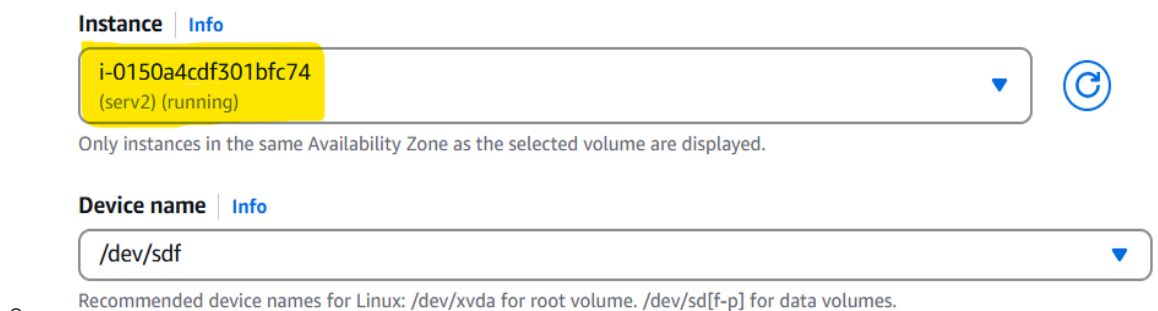


2. Attach Volume to New EC2 Instance:

- Click **Actions > Attach Volume**.



- Select the second EC2 instance to attach the volume to.



- Choose a device name (e.g., /dev/xvdf).

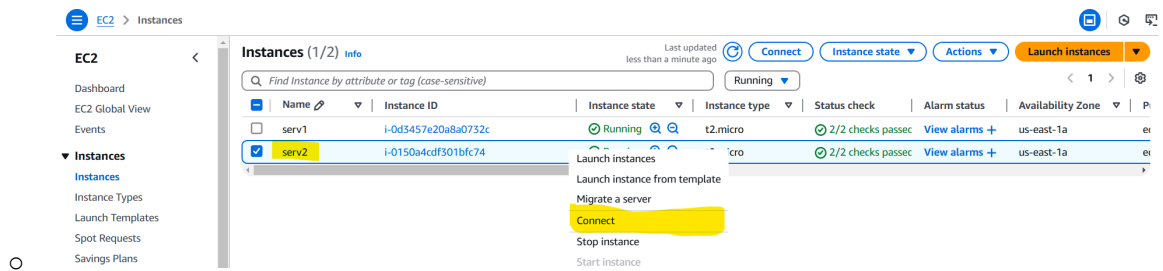
3. Complete Attachment:

- Click **Attach** to attach the volume to the new EC2 instance.

Step 8: Format and Mount the New Volume (on Second EC2 Instance)

1. Connect to the Second EC2 Instance:

- Use SSH to connect to the second EC2 instance where the new volume is attached.



2. Format the Volume:

- List the available disks using:
- Lsblk

```
[root@serv2 ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0   8G  0 disk
├─xvda1     202:1    0   8G  0 part /
├─xvda127   259:0    0   1M  0 part
├─xvda128   259:1    0  10M  0 part /boot/efi
└─xvdf      202:80   0   1G  0 disk
```

- Identify the new volume (e.g., /dev/xvdf), then format it:

```
[root@serv2 ~]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0   8G  0 disk
├─xvda1     202:1    0   8G  0 part /
├─xvda127   259:0    0   1M  0 part
├─xvda128   259:1    0  10M  0 part /boot/efi
└─xvdf      202:80   0   1G  0 disk
```

- sudo mkfs -t ext4 /dev/xvdf

3. Create Mount Point:

- Create a directory to mount the new volume:
- sudo mkdir /new_data

4. Mount the Volume:

- Mount the volume to the new directory:
- sudo mount /dev/xvdf /new_data

```
[root@serv2 bhanu]# ll
total 24
drwxr-xr-x. 2 root root 4096 Feb 24 18:13 dir1
drwxr-xr-x. 2 root root 4096 Feb 24 18:13 dir2
-rw-r--r--. 1 root root    0 Feb 24 18:12 file1
-rw-r--r--. 1 root root    0 Feb 24 18:12 file2
-rw-r--r--. 1 root root    0 Feb 24 18:12 file3
drwx-----. 2 root root 16384 Feb 24 18:09 lost+found
```

Final Notes:

- **Snapshots** are useful for backups or creating identical copies of volumes for different instances.
 - **Snapshots** are incremental, meaning only the changed blocks are saved, which helps reduce storage costs.
 - Always ensure the volume and EC2 instance are in the same **Availability Zone** when attaching volumes.
 - For regular data persistence, consider automating the backup and snapshot process.
-